Choice as an Act.

Russell's socks, cardinals as becomings, and the productive continuum,
machine-checked on the empty axiom list

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Supersedes v1.0.0 in §2 only: the sock impossibility is restated so that its load-bearing premise is formal rather than editorial, and a shoe is exhibited that the strengthened statement correctly spares.
Lean 4 corpus: `github.com/inventor1975/VRCycle` (tag `choice-v1.0.0`)
Verify from zero: see §8 — one file, one command, no dependencies

Abstract

Three classical set-theoretic themes — the axiom of choice on indistinguishable pairs (Russell's socks), the comparison of infinite cardinals, and the uncountability of the continuum — are re-read operationally: an assertion counts only as an *act*, performed and witnessed, never as a completed object postulated into existence. Under this reading each theme splits cleanly in two, and both halves become short machine-checked theorems. For the socks: no selection *rule* exists — no selector is at once blind to our labels and equivariant under the swap, and the two premises are formal, so the statement separates socks from shoes rather than proving the same of both; it is the symmetry argument with its premises exhibited, not the Fraenkel–Mostowski model-theoretic fact, which is cited and not formalised. Selection *acts*, meanwhile, form a continuum (the selectors are exactly the branches, which are not enumerable). For cardinals: a comparison is an act whose witness is data — an explicit injection $\mathbb{N} \hookrightarrow 2^{\mathbb{N}}$ is performed; the Cantor–Lawvere diagonal is proved uniformly for every floor of the power-set ladder, on the empty axiom list; the resulting order is *partial by design*, since cardinal trichotomy is equivalent to full AC and is cited as a formal-register label rather than claimed. For uncountability: the sign is flipped from prohibition to *productivity* — the fugitive from any enumeration is computed by an explicit term, so the continuum is productive in Post's sense: the catalogue that reads itself extends itself. Every proof is choice-free; the central results carry the empty axiom list, verified by `#print axioms`. Nothing here is a new classical theorem; the contribution is the operational re-reading, the axiom pricing, and the machine check.

The paper is written to be *verified from zero*: Section 8 gives a single self-contained file (no mathlib, no imports) that reproves every empty-axiom result in under a second, plus the full-corpus path and one audit table, so that any agent — human or machine — can re-derive every claim through the Lean 4 kernel without trusting the author or this text. An empty axiom list is precisely a verdict two parties who share no axioms and no trust can both confirm: the strongest form of a checkable claim.

1 Introduction: act against object

The VR cycle reads mathematics operationally: an object is what an operation reveals, an identity is earned by a witness, and nothing is asserted on credit. This note applies that discipline to the corner of set theory where credit is most concentrated: choice, cardinal comparison, and uncountability.

The organizing distinction is grammatical. A *rule* is a uniform prescription that could be written down once and applied everywhere; an *act* is a performance, individuated by its occurrence, whose

witness exists after and because of it. Classical set theory does not mark this distinction: the axiom of choice postulates a completed system of selections (an object), cardinal comparability postulates completed comparisons, and uncountability is stated as a prohibition. The operational reading marks the distinction everywhere, and the three classical themes then factor as follows.

- **Choice.** Where a rule exists, choice is available choice-free (dependent choice as recursion on a history-dependent rule; already in the cycle). Where a rule provably cannot exist — the indistinguishable pairs — selection survives as an act; and acts are plentiful precisely where rules are impossible (Section 2).
- **Cardinals.** A comparison of infinite types is an act: its witness (an injection) is data. Performed comparisons yield a *partial* order; the missing totality is not an oversight but a price tag, since trichotomy of cardinals is equivalent to full AC (Hartogs). The ladder of power floors never closes from below, by a uniform diagonal (Section 3).
- **Uncountability.** Read operationally it is not a wall but a generator: from every enumeration the escape is *computed*. The continuum is productive (Section 4).

What is not claimed. No classical theorem is contested, and none is new: the mathematical content of each proof is elementary and classical. The axiom of choice is not refuted — a symmetric-selector impossibility is a statement about *rules*, while AC postulates an object exempt from symmetry. Independence results (that ZF+DC proves neither AC nor the Banach–Tarski decomposition; Solovay's model) are metatheory: cited, never formalised here. The contribution is the re-reading, the split of each theme into an impossible-rule half and a performed-act half, the axiom pricing of every step, and the machine check.

The instrument. Everything is formalised over the cycle's becoming-space: $\mathsf{Branch} := \mathbb{N} \to \mathsf{Bool}$, the space of performed binary sequences, with its finite performed segments (`Branch.take`). The corpus is Lean 4 over mathlib; every public object ends with `#print axioms`. "Empty axiom list" below means exactly that: not even `propext`.

2 Russell's socks: rules number zero, acts a continuum

A countable family of pairs of indistinguishable socks admits no choice rule: this is Russell's classical illustration, whose set-theoretic home is the Fraenkel–Mostowski permutation models. The operational rendering: the two socks of box k are `false`/`true` only in the formal register's bookkeeping; the box itself offers no mark, so the *swap* at box k is a symmetry of the situation.

Theorem 2.1 (`no_symmetric_selector`; empty axiom list)**.** *No selection is swap-invariant beyond any finite bound: for every selector $c : \mathbb{N} \to \mathsf{Bool}$ and every bound N there is a box $k \geq N$ at which the swap moves the selection.*

The proof is one line — the swap at k flips the pick at k, and no Boolean equals its own negation.

That line is too cheap to carry the claim, and saying so is the point. Read as *"therefore no rule exists"*, the theorem above proves too much: nothing in its statement mentions indistinguishability, so the same text goes through for *shoes*, where a rule plainly does exist. What separates the two cases — that the swap is a symmetry of the sock situation and not of the shoe situation — was carried by the prose of the preceding paragraph, not by the formalisation. A machine-checked

claim whose load-bearing premise is editorial is exactly the failure this cycle exists to catch, so it is corrected here rather than defended.

Two definitions move the premise into the statement. A *labelling* $l : \mathbb{N} \to \mathsf{Bool}$ records which element of each pair we have named first; a *selector* s reads a labelling and picks in every box.

Definition 2.2 (the two premises, now formal)**.** Equivariant$(s) := \forall l\, k,\ s(\mathrm{swap}_k l)(k) = \neg\, s(l)(k)$ *— renaming the two elements of pair k carries the pick with them, because it is the same object under a different name. This is what "indistinguishable" means operationally.*
Blind$(s) := \forall l\, l'\, k,\ s(l)(k) = s(l')(k)$ *— the box offers no mark, so no labelling of ours can enter the choice.*

Theorem 2.3 (`socks_no_rule`; empty axiom list)**.** *No selector is both blind and equivariant.*

Theorem 2.4 (`shoes_rule_exists`, `shoes_not_blind`; empty axiom list)**.** leftShoe $:= \lambda l\, k.\, l(k)$ *is equivariant and is not blind. A rule for shoes therefore exists and the impossibility does not touch it.*

The second theorem is not decoration: without it the first would be an impossibility proved about nothing in particular. Together they say where the obstruction lives — in the collision of blindness with equivariance, and nowhere else. The depth is unchanged; the *discrimination* is what was missing.

What this is and is not. It is a machine-checked formalisation of the symmetry argument, with its premises exhibited. It is *not* Fraenkel–Mostowski: that the family genuinely has no choice function in a permutation model is a statement about models, which is cited here and formalised nowhere in this note. The reader who wants the model-theoretic fact should go to Fraenkel and Mostowski; what is offered here is the operational grammar of the same obstruction.

What the classical discussion obscures is the other half:

Theorem 2.5 (`selectors_not_enumerable`; empty axiom list)**.** *The selectors are exactly the branches; hence they are not enumerable: no map $\mathbb{N} \to (\mathbb{N} \to \mathsf{Bool})$ is surjective.*

Selection over indistinguishable pairs therefore exists *never as a rule and uncountably as an act*: any performed sequence — operationally, a lawless coin — selects. Rules: zero. Acts: a continuum.

Remark 2.6 (The computational face). The same split is a classical fact of distributed computing: deterministic symmetric agents cannot break symmetry (leader election in anonymous networks is deterministically impossible; Angluin 1980, Itai–Rodeh 1981), while randomized agents succeed almost surely in expected two rounds. A companion simulation in the ZTL workspace measures exactly this: deterministic twins never disagree; coin-equipped socks break symmetry in ≈ 2 rounds. The deterministic half is itself a theorem of the corpus (`twins_never_break`, `no_unique_leader`; empty axiom list): in an anonymous network of identical automata started identically, the configuration is constant across nodes at every round *for arbitrary wiring* — stronger than a ring — so no round distinguishes a unique node under any predicate. To our knowledge this folklore impossibility had not been machine-checked before. The sock chooses itself, or nothing chooses it.

3 Cardinals as becomings

Definition 3.1 (Witnessed comparison)**.** OpInj$(A, B) := \{f : A \to B \mid f$ *injective*$\}$*, carried as* data*: a performed comparison is an explicit injection, not an existence claim.*

Proposition 3.2 (`natIntoBranch`; empty axiom list)**.** *An explicit injection* $\mathbb{N} \hookrightarrow \mathsf{Branch}$ *is performed:* $n \mapsto$ *the branch that fires exactly at* n*.*

Theorem 3.3 (`cantor_ladder`; empty axiom list)**.** *For every type* A*, no map* $A \to (A \to \mathsf{Bool})$ *is surjective.*

The diagonal is stated once and holds at *every* floor: each doubling — each added colleague — opens a floor the previous one cannot reach. Combining the two:

Corollary 3.4 (`nat_strictly_below_branch`; empty axiom list)**.** $\mathbb{N}$ *is operationally strictly below its power floor: the injection up is performed, and no surjection comes back.*

The order is partial by design. Performed comparisons compose, but no totality is claimed: trichotomy of cardinals is equivalent to the full axiom of choice (Hartogs), i.e. to the completed object this note declines. Two becomings are comparable where a comparison has been performed — and nowhere else for free. The missing comparisons are not unknown truths; they are unperformed acts.

4 Productive uncountability: the sign flipped

Classically, uncountability is a prohibition: *there is no* enumeration. Operationally the same content is a production:

Definition 4.1 (`escape`)**.** $\mathrm{escape}(e) := n \mapsto \neg\, e(n)(n)$*, for any* $e : \mathbb{N} \to \mathsf{Branch}$*.*

Theorem 4.2 (`escape_escapes`, `branches_productive`; empty axiom list)**.** *For every enumeration* e *and every* k*,* $\mathrm{escape}(e) \neq e(k)$*; hence for every enumeration there* is *a branch it misses, and the witness is handed over, not merely asserted.*

The fugitive is a *term*: computed from the catalogue, differing from each entry at that entry's own line. This is productivity in Post's sense, transplanted from computability theory to the becoming-space: the continuum is not a large warehouse but a mirror — *the catalogue that reads itself extends itself.* Uncountability is thereby re-signed from a wall to a generator: every attempt to complete the listing manufactures the next item. (In the cycle's earlier ZTL expedition the same theme split into non-registrability and incompleteness of registries, measured over marked streams; the present theorem is the choice-free kernel of that split.)

5 The demarcation: DC and the writable remainder

The cycle already contains the positive half of choice: a history-dependent rule determines a branch by recursion, choice-free (`operational_dependent_choice`) — dependent choice is the *performable* part of AC, the part with process semantics: each selection is made in the context of those already made, and a witness accrues step by step. The present note adds the negative half (no rule on symmetric pairs) and the act half (any branch selects). What remains of full AC above DC is precisely the part that can only be *written*: simultaneous selection over families no process traverses. That remainder buys the familiar exotica — and, by Solovay's model (metatheory, cited), giving it up costs analysis nothing while dissolving, e.g., the Banach–Tarski decomposition, whose non-measurable pieces are choice-certificates that can never be exhibited. In the cycle's terms: DC is choice earned in acts; the remainder is choice on permanent credit.

6 Findings

Finding 6.1. **Rules zero, acts a continuum.** Selection over indistinguishable pairs is impossible as a rule (empty axiom list) and uncountably plentiful as an act. The two halves are one theorem apart in the same file.

Finding 6.2. **Comparison is an act; the order is partial by design.** Trichotomy is not lost — it is priced: it costs exactly full AC, and the operational register declines the purchase.

Finding 6.3. **The ladder never closes from below**, uniformly and on the empty axiom list: one diagonal serves every floor.

Finding 6.4. **Uncountability is a productivity.** The positive form — witness handed over — is strictly more informative than the classical prohibition, and costs no axioms at all.

Finding 6.5. **The premise confession** (new in v1.0.1). In v1.0.0 the sock impossibility was machine-checked, and what the machine checked was that no Boolean equals its own negation. The premise that made it about *socks* — that the swap is a symmetry of that situation and not of the shoe situation — sat in the prose. The diagnostic is blunt and worth keeping: *the theorem could not tell socks from shoes.* It is repaired by moving both premises into the statement (`socks_no_rule`) and exhibiting the case the statement must spare (`shoes_rule_exists`, `shoes_not_blind`); the mathematics is no deeper, the discrimination is what was added. The lesson generalises past this paper: an empty axiom list certifies the step from premises to conclusion, and certifies *nothing* about whether the premises say what the surrounding sentences claim. A formalisation that proves the same theorem about the case it means to exclude has not yet formalised the distinction it is about.

Finding 6.6. **The tier confession.** The cycle's central diagonal (`branches_not_enumerable`) had carried `[propext]` since its first proof — inherited not from the library but from our own `simp` on a Boolean hypothesis. One explicit `Bool.noConfusion` restored the empty list. The lesson is methodological: before blaming the substrate, audit your own tactics. MEASURED: the current audit (Table 1) shows the empty list for every theorem named in this note save one — DC, which carries `[propext]`, marked and not load-bearing for the thesis.

7 Relation to the literature

Fraenkel's and Mostowski's permutation models are the classical home of the sock impossibility, and remain its only home here: this note formalises the *symmetry argument* over selections — blindness against equivariance, with a shoe exhibited to show the statement discriminates — and does not formalise the model-theoretic fact. The distinction is stated because v1.0.0 blurred it. Post's productive sets supply the name and the shape of Section 4. Angluin's and Itai–Rodeh's results on anonymous leader election are the computational face of the rule/act split. The diagonal in the uniform form is Lawvere's fixed-point argument specialised to Bool. Brouwer's choice sequences stand behind the branch space and the lawless coin. None of these results is claimed as new; the operational re-reading, the splitting of each theme into rule-half and act-half, and the empty-list machine check are the contribution.

8 Machine verification: check it from zero

This section is written so that any agent — human or machine — can verify every claim of this paper *without trusting the author, the AI that helped write it, or this text.* That is the point of the

empty axiom list. A theorem whose `#print axioms` is empty is provable from nothing: two parties who share no axioms, no training, and no trust can still both compute the same verdict, because the Lean 4 kernel re-derives it from zero. This is the strongest form of a zero-trust claim, and it is mechanical, not rhetorical.

The fast path: one file, one command, no dependencies. Every empty-axiom-list theorem of this paper is reproved, from scratch, in a single self-contained file `Verify_Choice_standalone.lean` at the repository root — *no mathlib, no lake, no imports.* Any agent checks the lot in under a second:

```
lean Verify_Choice_standalone.lean
```

It prints, for each of the thirteen theorems, `does not depend on any axioms`. This is the whole zero-trust claim in executable form: the bare Lean 4 kernel (version in `lean-toolchain`, 4.29.1) re-derives every verdict from nothing — no library to trust, no author to trust, no text to trust. Two parties who share only the kernel can both run it and agree.

The full path (with the substrate).

```
git clone https://github.com/inventor1975/VRCycle
cd VRCycle          # tag choice-v1.0.0; toolchain in lean-toolchain (Lean 4.29.1)
lake exe cache get  # mathlib cache
lake build          # verifies the whole corpus (~8370 jobs)
```

The `#print axioms` lines at the foot of `VRCycle/Continuum/{Choice,Cardinal,Branch}.lean` then print, for every named theorem, exactly the axiom profile tabulated below. A verifier compares the printed lines against Table 1; any mismatch falsifies the paper. Nothing is taken on the author's word. (The standalone file and the in-corpus versions prove the same statements; the corpus adds the operational framing and the `[propext]`-carrying DC theorem, which the standalone omits since it is not empty-list.)

Theorem	Axioms	Claim
`no_symmetric_selector`	[]	no swap-symmetric selection rule
`selectors_not_enumerable`	[]	the selection acts are a continuum
`twins_never_break`	[]	anonymous determinism keeps symmetry
`no_unique_leader`	[]	hence no round distinguishes a node
`natIntoBranch`	[]	$\mathbb{N} \hookrightarrow$ Branch performed
`cantor_ladder`	[]	no floor surjects onto its power
`nat_strictly_below_branch`	[]	$\mathbb{N}$ strictly below $2^{\mathbb{N}}$
`escape_escapes`	[]	the fugitive differs from every entry
`branches_productive`	[]	uncountability with witness handed over
`branches_not_enumerable`	[]	the diagonal (tier-cleaned)
`operational_dependent_choice`	`[propext]`	DC as recursion on a rule

Table 1: The audit table. "[]" = the empty axiom list: provable from nothing, not even `propext`. Every row is a line the build prints; a verifier confirms each independently. The single `[propext]` is honest and marked — it is the one place a proof-irrelevance rewrite was used, and it is not load-bearing for the paper's thesis (the impossibility and the diagonal both stand at []).

The companion simulations (symmetry breaking, the quantum coin) live in the ZTL workspace (`github.com/inventor1975/ZTL`, `dilemmas/`); they are evidence, not proof, and are labelled as such — only the Lean theorems above carry the empty-axiom guarantee.

Acknowledgement

This work was carried out with the substantial participation of the AI system Claude (Anthropic; this draft — Claude Fable 5) in a dialogue setting; all design decisions, fork choices, and final responsibility rest with the human author. The reliability of the results does not depend on trusting the AI: every claim is checkable by the Lean 4 kernel and the repository's audit lines.